Complete CI/CD Mastery with GitHub Actions

1. CI Pipeline using GitHub Actions with Docker (Build Verification)

```
Command Prompt
Microsoft Windows [Version 10.0.26200.8117]
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C:\Users\sreev>mkdir my-docker-app

C:\Users\sreev>cd my-docker-app

C:\Users\sreev\my-docker-app>mkdir .github

C:\Users\sreev\my-docker-app>dir
 Volume in drive C is OS
 Volume Serial Number is 0CE8-3131

 Directory of C:\Users\sreev\my-docker-app

02-04-2026  14:24    <DIR>          .
02-04-2026  14:24    <DIR>          ..
02-04-2026  14:24    <DIR>          .github
               0 File(s)              0 bytes
               3 Dir(s)  42,354,270,208 bytes free

C:\Users\sreev\my-docker-app>notepad app.py
```
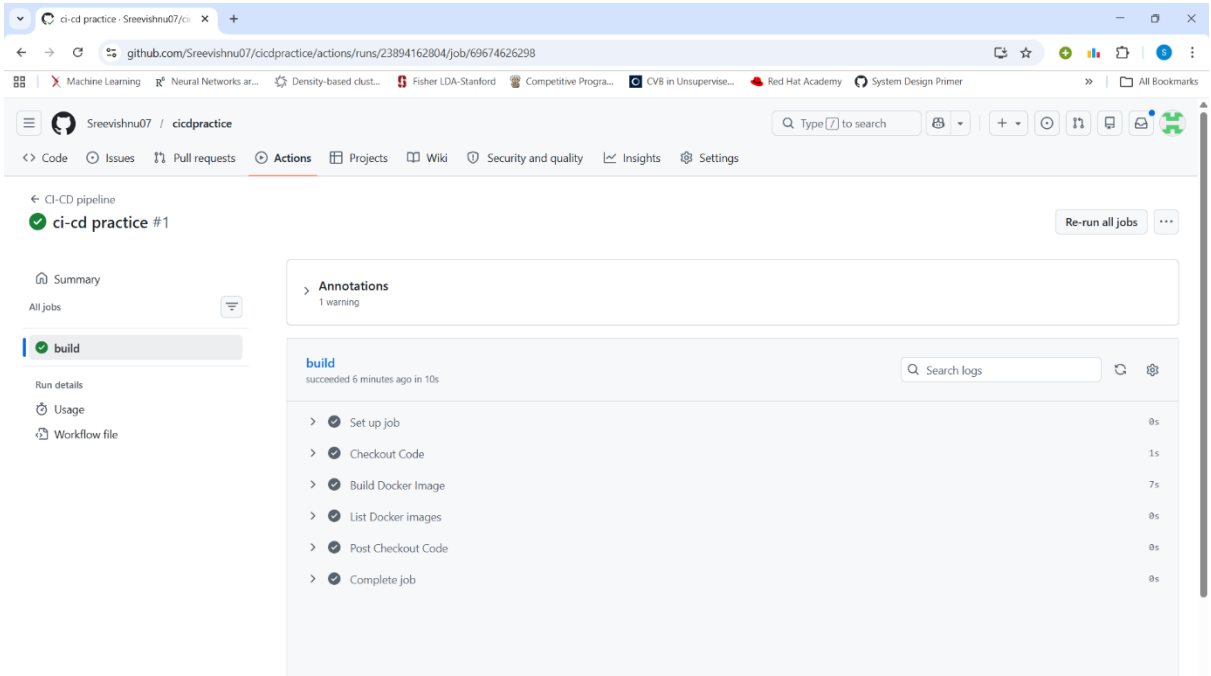

```
Command Prompt

C:\Users\sreev\my-docker-app>dir
 Volume in drive C is OS
 Volume Serial Number is 0CE8-3131

 Directory of C:\Users\sreev\my-docker-app

02-04-2026  14:53    <DIR>          .
02-04-2026  14:24    <DIR>          ..
02-04-2026  14:24    <DIR>          .github
02-04-2026  14:29               171 app.py
02-04-2026  14:52               108 dockerfile
02-04-2026  14:31                 5 requirements.txt
               3 File(s)            284 bytes
               3 Dir(s)  39,422,799,872 bytes free

C:\Users\sreev\my-docker-app>cd .github

C:\Users\sreev\my-docker-app\.github>dir
 Volume in drive C is OS
 Volume Serial Number is 0CE8-3131

 Directory of C:\Users\sreev\my-docker-app\.github

02-04-2026  14:24    <DIR>          .
02-04-2026  14:53    <DIR>          ..
               0 File(s)              0 bytes
               2 Dir(s)  39,385,759,744 bytes free

C:\Users\sreev\my-docker-app\.github>mkdir workflows
```

```
Command Prompt
C:\Users\sreev\my-docker-app>git init
Initialized empty Git repository in C:/Users/sreev/my-docker-app/.git/

C:\Users\sreev\my-docker-app>git remote add origin git@github.com:Sreevishnu07/cicdpractice.git

C:\Users\sreev\my-docker-app>git add .

C:\Users\sreev\my-docker-app>git commit -m "ci-cd practice"
[master (root-commit) 532531d] ci-cd practice
 4 files changed, 30 insertions(+)
 create mode 100644 .github/workflows/ci-cd.yml
 create mode 100644 app.py
 create mode 100644 dockerfile
 create mode 100644 requirements.txt
```

```
C:\Users\sreev\my-docker-app>git branch -M main

C:\Users\sreev\my-docker-app>git push -u origin  main
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 12 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (8/8), 865 bytes | 288.00 KiB/s, done.
Total 8 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:Sreevishnu07/cicdpractice.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

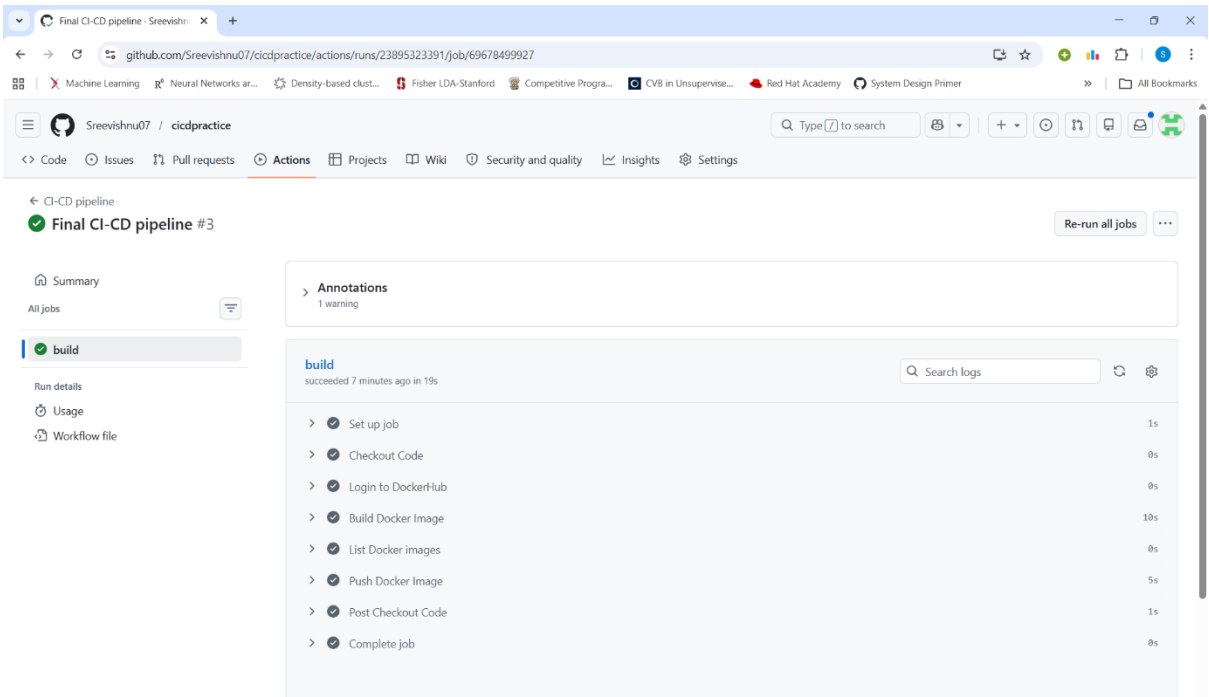



```
C:\Users\sreev\my-docker-app>type .github\workflows\ci-cd.yml
name: CI-CD pipeline
on:
  push:
    branches:
      - main
jobs:
  build:
    runs-on: ubuntu-latest
    steps:
      - name: Checkout Code
        uses: actions/checkout@v4

      - name: Build Docker Image
        run: docker build -t my-docker-app .

      - name: List Docker images
        run: docker images
```

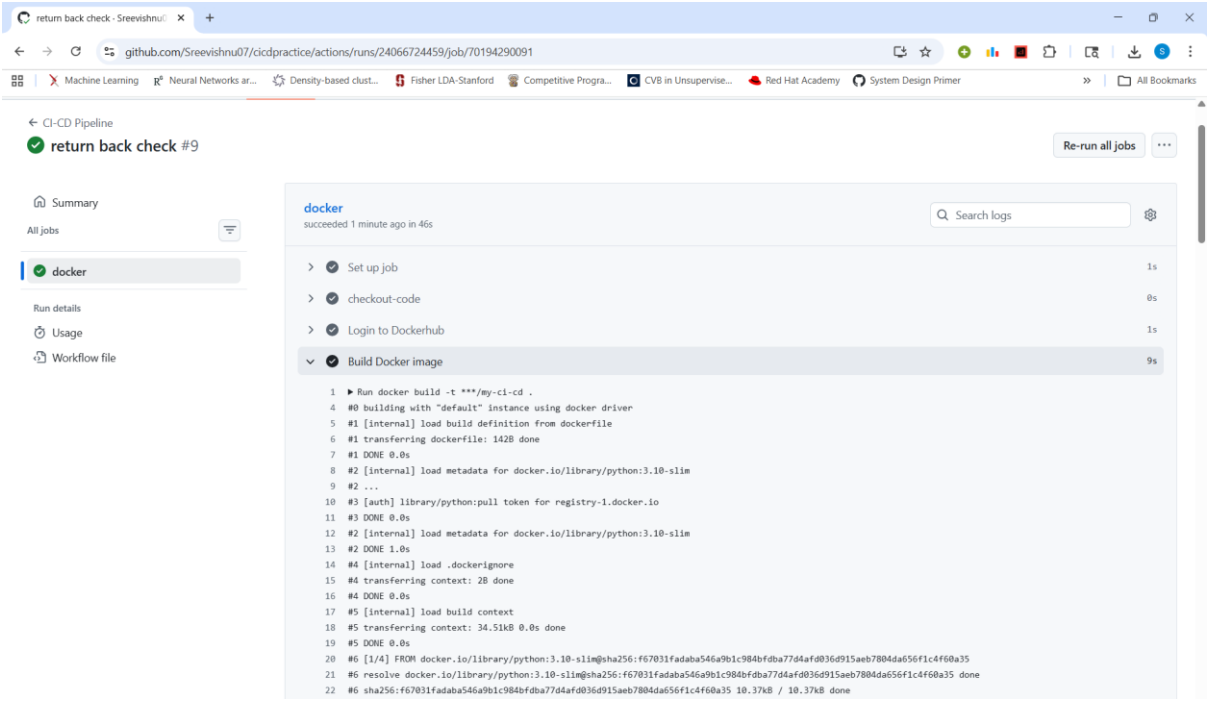
2. CI/CD Pipeline using GitHub Actions with DockerHub Integration

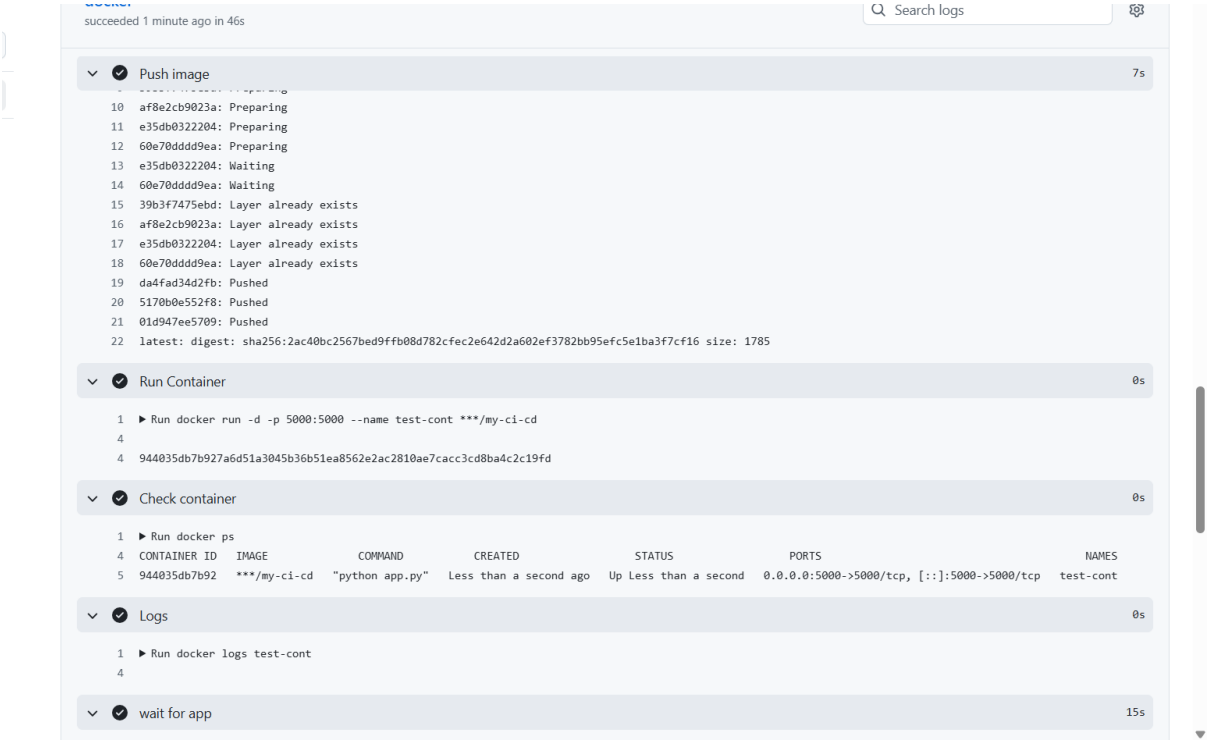
```
C:\Users\sreev\my-docker-app>notepad .github\workflows\ci-cd.yml

C:\Users\sreev\my-docker-app>git add .

C:\Users\sreev\my-docker-app>git commit -m "Final CI-CD pipeline"
[main 5ab8af8] Final CI-CD pipeline
 1 file changed, 1 insertion(+), 1 deletion(-)

C:\Users\sreev\my-docker-app>git push -u origin main
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (5/5), 502 bytes | 502.00 KiB/s, done.
Total 5 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:Sreevishnu07/cicdpractice.git
   c4cdf9b..5ab8af8  main -> main
branch 'main' set up to track 'origin/main'.
```

```
C:\Users\sreev\my-docker-app>type .github\workflows\ci-cd.yml
name: CI-CD pipeline
on:
  push:
    branches:
      - main
jobs:
  build:
    runs-on: ubuntu-latest
    steps:
      - name: Checkout Code
        uses: actions/checkout@v4

      - name: Login to DockerHub
        run: echo "${{secrets.DOCKER_PASSWORD}}" | docker login -u "${{secrets.DOCKER_USERNAME}}" --password-stdin

      - name: Build Docker Image
        run: docker build -t ${{secrets.DOCKER_USERNAME}}/my-docker-app .

      - name: List Docker images
        run: docker images

      - name: Push Docker Image
        run: docker push ${{secrets.DOCKER_USERNAME}}/my-docker-app
```



3. End-to-End CI/CD Pipeline for Containerized Flask Application using Docker and GitHub Actions





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Summary

All jobs

docker

Run details

Usage

Workflow file

docker
succeeded 1 minute ago in 46s

Search logs

wait for app 15s

test application 0s

```
▶ Run curl http://localhost:5000
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0
100    23  100    23    0     0  11420      0 --:--:-- --:--:-- --:--:-- 23000
Hello from DOCKER CI/CD
```

stop container 10s

```
▶ Run docker stop test-cont

test-cont
```

remove cont 0s

```
▶ Run docker rm test-cont
test-cont
```

Post checkout-code 0s

```
Post job cleanup.
/usr/bin/git version
git version 2.53.0
Temporarily overriding HOME='/home/runner/work/_temp/e9427330-6447-4ee7-b664-5f8040226cb6' before making global git config changes
Adding repository directory to the temporary git global config as a safe directory
/usr/bin/git config --global --add safe.directory /home/runner/work/cicdpractice/cicdpractice
/usr/bin/git config --local --name-only --get-regexp core\.sshCommand
/usr/bin/git submodule foreach --recursive sh -c "git config --local --name-only --get-regexp 'core\.sshCommand' && git config --local --unset-all
'core.sshCommand' || :"
/usr/bin/git config --local --name-only --get-regexp http\.https\:\/\/github\.com\/\.extraheader
```

cicdpractice/.github/workflows

github.com/Sreevishnu07/cicdpractice/blob/main/.github/workflows/ci-cd.yml

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Sreevishnu07 Update CI-CD workflow to simplify wait for app step

17a5a8b · 13 minutes ago History

Files

main

Go to file

.github/workflows

ci-cd.yml

app.py

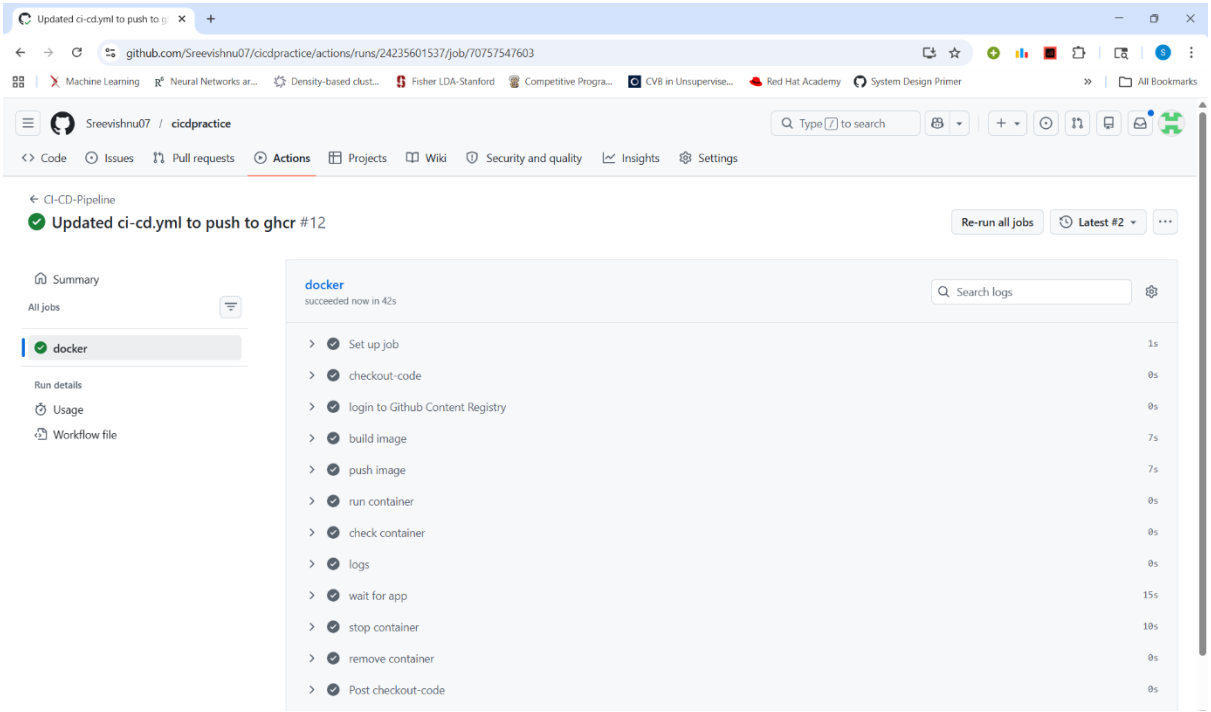
dockerfile

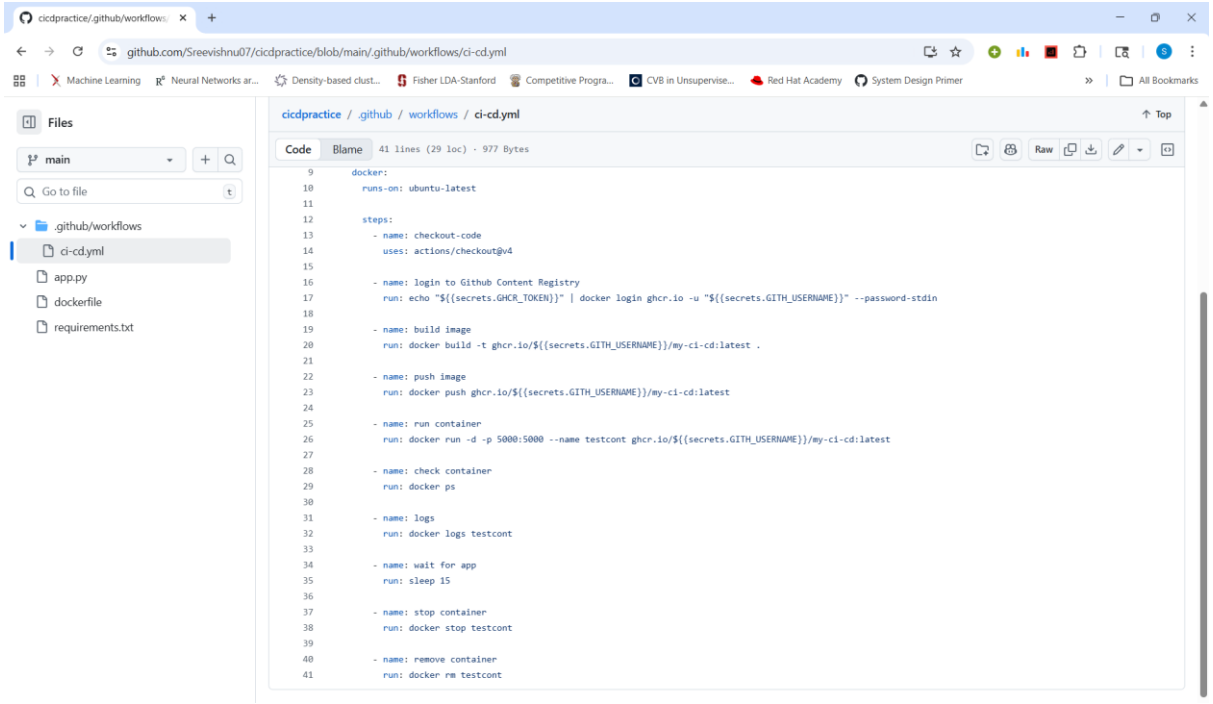
requirements.txt

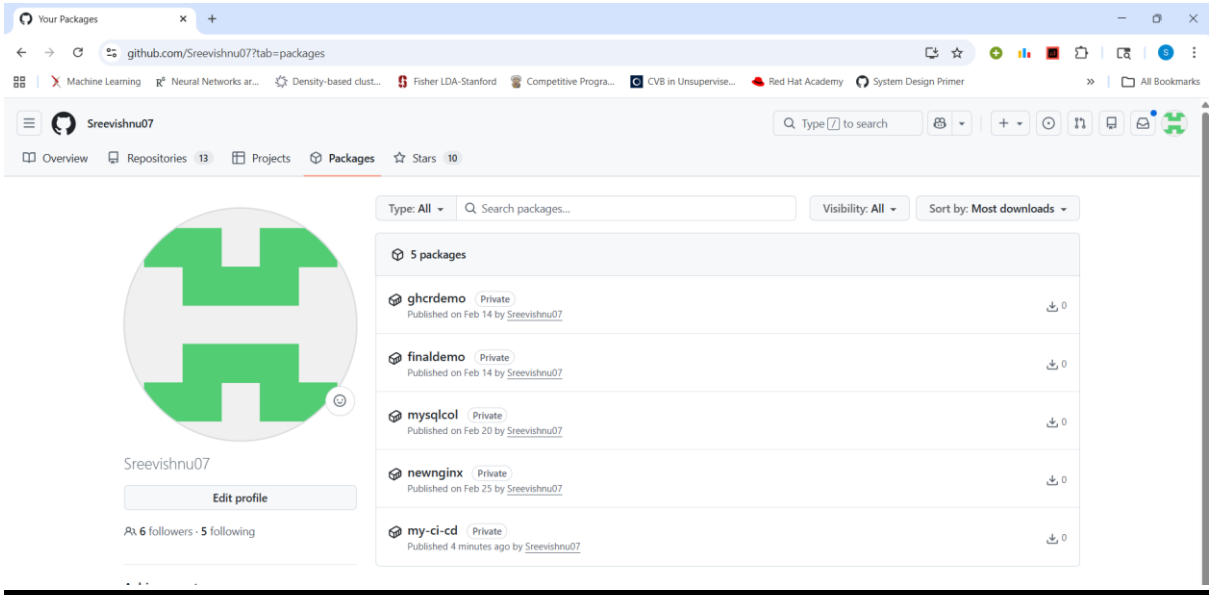
Code | Blame | 44 lines (31 loc) · 1010 Bytes | Raw

```yaml
name: CI-CD Pipeline

on:
  push:
    branches:
      - main

jobs:
  docker:
    runs-on: ubuntu-latest

    steps:
      - name: checkout-code
        uses: actions/checkout@v4

      - name: Login to Dockerhub
        run: echo "${{ secrets.DOCKER_PASSWORD }}" | docker login -u "${{ secrets.DOCKER_USERNAME }}" --password-stdin

      - name: Build Docker image
        run: docker build -t ${{ secrets.DOCKER_USERNAME }}/my-ci-cd .

      - name: Push image
        run: docker push ${{ secrets.DOCKER_USERNAME }}/my-ci-cd

      - name: Run Container
        run: docker run -d -p 5000:5000 --name test-cont ${{ secrets.DOCKER_USERNAME }}/my-ci-cd

      - name: Check container
        run: docker ps

      - name: Logs
        run: docker logs test-cont
```

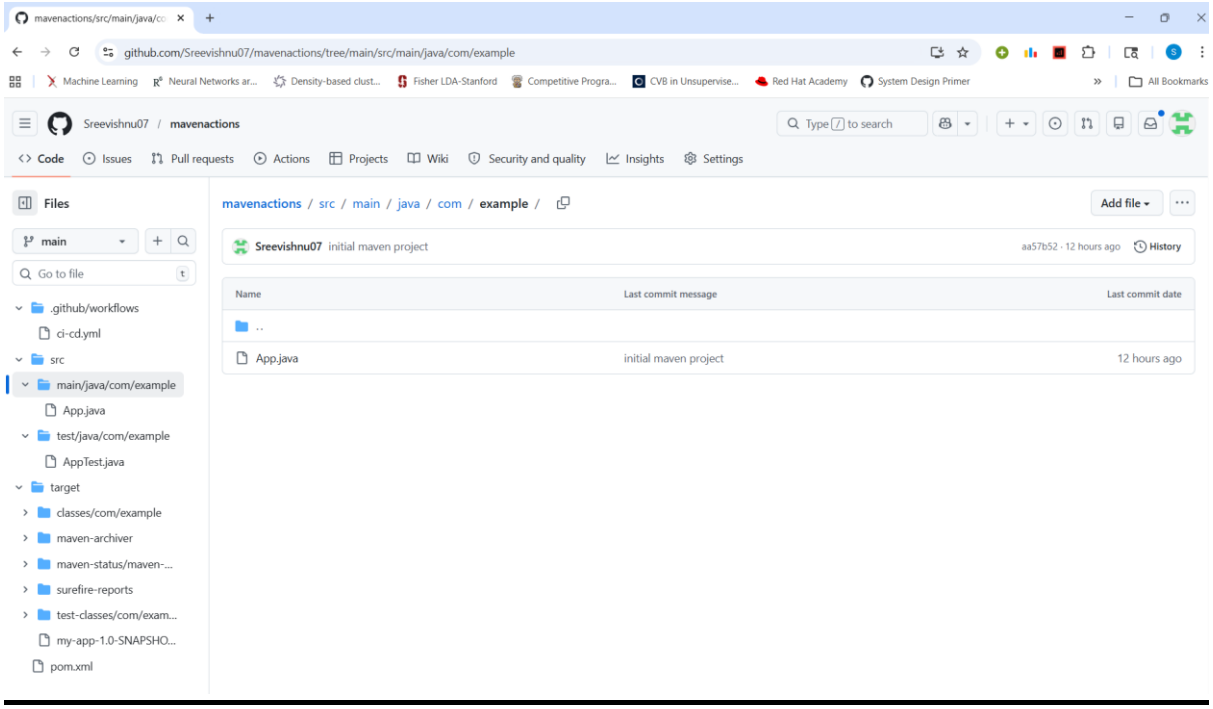
4. Automated CI/CD Pipeline for Docker Image Deployment using GHCR







5. CI Pipeline Implementation with Maven Build, Test Automation and Artifact Management via GitHub Actions


mavenactions/src/main/java/co
github.com/Sreevishnu07/mavenactions/tree/main/src/main/java/com/example
Machine Learning
Neural Networks ar...
Density-based clust...
Fisher LDA-Stanford
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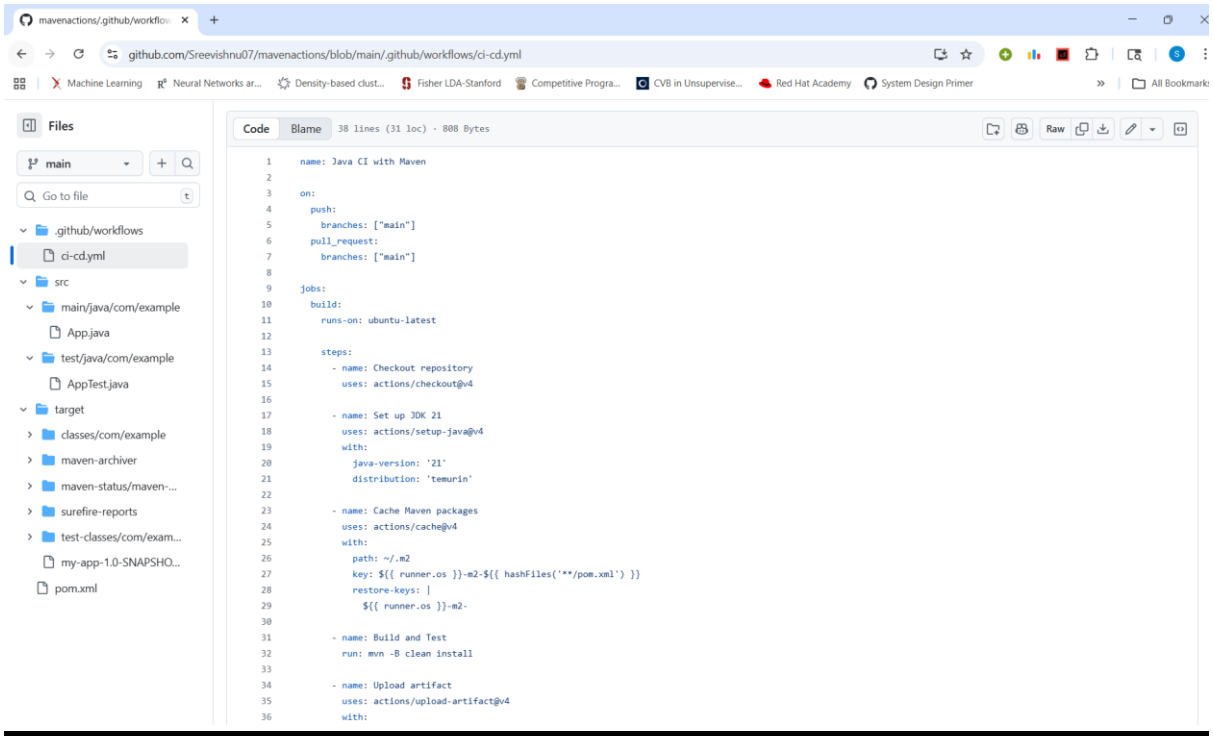
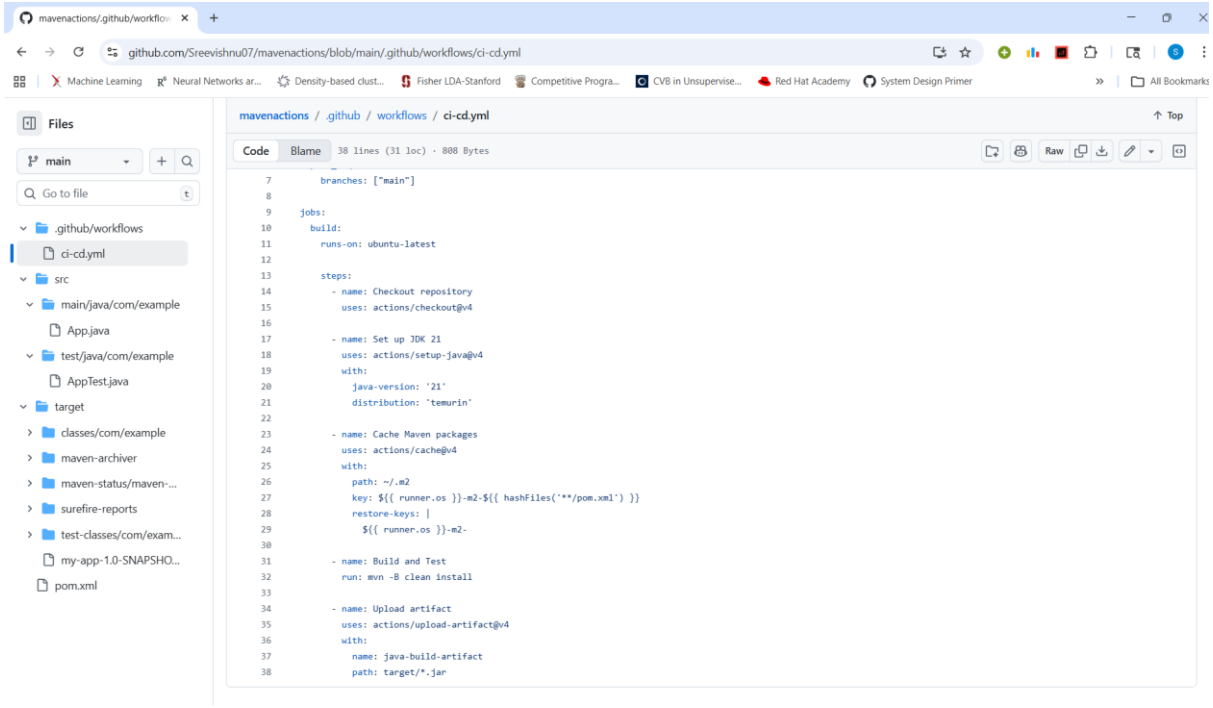
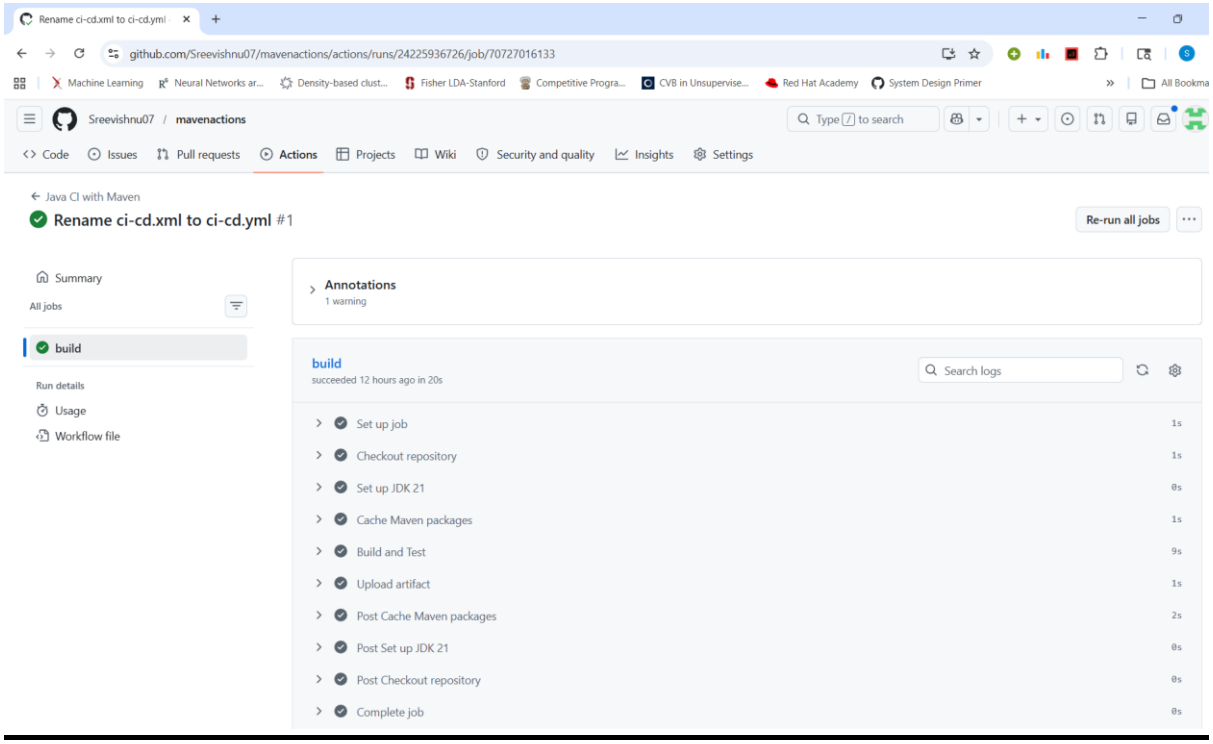
Code
Issues
Pull requests
Actions
Projects
Wiki
Security and quality
Insights
Settings
Files
main
Go to file
.github/workflows
ci-cd.yml
src
main/java/com/example
App.java
test/java/com/example
AppTest.java
target
classes/com/example
maven-archiver
maven-status/maven-...
surefire-reports
test-classes/com/exam...
my-app-1.0-SNAPSHO...
pom.xml
mavenactions / src / main / java / com / example /
Add file
Sreevishnu07 initial maven project
aa57b52 · 12 hours ago
History
Name
Last commit message
Last commit date
..
App.java
initial maven project
12 hours ago


mavenactions/.github/workflo
github.com/Sreevishnu07/mavenactions/blob/main/.github/workflows/ci-cd.yml
Machine Learning
Neural Networks ar...
Density-based clust...
Fisher LDA-Stanford
Competitive Progra...
CVB in Unsupervise...
Red Hat Academy
System Design Primer
All Bookmarks
Files
main
Go to file
.github/workflows
ci-cd.yml
src
main/java/com/example
App.java
test/java/com/example
AppTest.java
target
classes/com/example
maven-archiver
maven-status/maven-...
surefire-reports
test-classes/com/exam...
my-app-1.0-SNAPSHO...
pom.xml
Code
Blame
38 lines (31 loc) · 808 Bytes
Raw
name: Java CI with Maven
on:
push:
branches: ["main"]
pull_request:
branches: ["main"]
jobs:
build:
runs-on: ubuntu-latest
steps:
- name: Checkout repository
uses: actions/checkout@v4
- name: Set up JDK 21
uses: actions/setup-java@v4
with:
java-version: '21'
distribution: 'temurin'
- name: Cache Maven packages
uses: actions/cache@v4
with:
path: ~/.m2
key: ${{ runner.os }}-m2-${{ hashFiles('**/pom.xml') }}
restore-keys: |
${{ runner.os }}-m2-
- name: Build and Test
run: mvn -B clean install
- name: Upload artifact
uses: actions/upload-artifact@v4
with:


mavenactions/.github/workflows
github.com/Sreevishnu07/mavenactions/blob/main/.github/workflows/ci-cd.yml
Machine Learning
Neural Networks ar...
Density-based clust...
Fisher LDA-Stanford
Competitive Progra...
CVB in Unsupervise...
Red Hat Academy
System Design Primer
All Bookmarks
Files
main
Go to file
.github/workflows
ci-cd.yml
src
main/java/com/example
App.java
test/java/com/example
AppTest.java
target
classes/com/example
maven-archiver
maven-status/maven-...
surefire-reports
test-classes/com/exam...
my-app-1.0-SNAPSHO...
pom.xml
mavenactions / .github / workflows / ci-cd.yml
Top
Code
Blame
38 lines (31 loc) · 808 Bytes
Raw
branches: ["main"]
jobs:
build:
runs-on: ubuntu-latest
steps:
- name: Checkout repository
uses: actions/checkout@v4
- name: Set up JDK 21
uses: actions/setup-java@v4
with:
java-version: '21'
distribution: 'temurin'
- name: Cache Maven packages
uses: actions/cache@v4
with:
path: ~/.m2
key: ${{ runner.os }}-m2-${{ hashFiles('**/pom.xml') }}
restore-keys: |
${{ runner.os }}-m2-
- name: Build and Test
run: mvn -B clean install
- name: Upload artifact
uses: actions/upload-artifact@v4
with:
name: java-build-artifact
path: target/*.jar


Rename ci-cd.xml to ci-cd.yml
github.com/Sreevishnu07/mavenactions/actions/runs/24225936726/job/70727016133
Machine Learning
Neural Networks ar...
Density-based clust...
Fisher LDA-Stanford
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Code
Issues
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Security and quality
Insights
Settings
Java CI with Maven
Rename ci-cd.xml to ci-cd.yml #1
Re-run all jobs
Summary
All jobs
build
Run details
Usage
Workflow file
Annotations
1 warning
build
succeeded 12 hours ago in 20s
Search logs
Set up job 1s
Checkout repository 1s
Set up JDK 21 0s
Cache Maven packages 1s
Build and Test 9s
Upload artifact 1s
Post Cache Maven packages 2s
Post Set up JDK 21 0s
Post Checkout repository 0s
Complete job 0s